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EN2160: Electronic Design Realization

Digital Communication Visualization Console

An Interactive Embedded System for Visualizing
Fundamental Concepts of Digital Communication Systems

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Abstract

Abstract mathematical paradigms in telecommunication engineering, including variable-length source entropy reductions, linear block cyclic redundancy protections, and complex constellation symbol mappings, present significant pedagogical barriers in traditional undergraduate lecturing formats. This report documents the complete design realization of the *Digital Communication Visualization Console* by Team Synex for module EN2160. Driven by a high-performance ARM Cortex-M based STM32 microcontroller, this hardware instrument provides a localized physical verification loop for abstract concepts. Processing a user-defined alphabet set ($S = \{A, B, C, D, E\}$), the console calculates and renders multi-stage transformations—from Source Coding (Fixed-length, Huffman, Shannon-Fano) to Packetization framing configurations, Channel Coding (Parity, CRC, Hamming), and digital modulations (ASK, BPSK, QPSK, QAM). Featuring twin 3.5” SPI TFT panels managed via Direct Memory Access (DMA), dynamic 74HC595 shift register LED monitoring arrays, and physical potentiometer-driven additive noise emulation circuits interfacing with internal 12-bit ADCs, the device allows students to visually audit transmission metrics, code-word transitions, and real-time constellation degradation.

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1 Introduction

1.1 Background

Modern telecommunication systems rely on multi stage digital signal manipulation architectures to maximize data transfer rates, minimize operational bandwidth utilization, and defend information vectors against channel impairments. In pedagogical frameworks, these multi stage blocks are fundamentally decoupled into strict mathematical abstractions: informational entropy metrics, orthogonal phase space projections. While critical for core engineering theory, these mathematical concepts frequently present comprehension challenges for undergraduate students when separated from tactile, real time hardware manifestations.

1.2 Motivation

The inspiration for this design is rooted in the current educational gap within electrical and telecommunication laboratory courses in Sri Lanka. Engineering students typically evaluate communication pipelines via pure software simulation scripts (e.g., MATLAB or Python environments). Although software environments offer numerical flexibility, they separate students from physical timing restrictions, real time clock domains, and hardware interface realities. This project establishes an interactive, intermediate educational desktop platform that delivers high fidelity visual and tactile feedback in real time.

1.3 Problem Statement

Digital communication is a cornerstone domain within electrical, electronic, and computer engineering curricula. However, undergraduate engineering students often master the underlying abstract mathematical frameworks while struggling to build a clear mental model of physical realization. Specifically, students struggle to visualize:

- The operational behavior and code-tree generation variations of dynamic variable-length source compilers.
- How bitstreams are partitioned into distinct boundaries to execute packetization workflows.
- How redundancy bits (CRC or parity) are mathematically appended to message frames.
- The explicit mechanism by which random channel noise corrupts transmission bits.
- How corrupted packets are evaluated, flags are trigged, and bad frames are ultimately rejected and discarded by receiver logic.

1.4 Stakeholder Analysis

To map the deployment environment effectively, a complete stakeholder analysis was executed:

- **Primary Stakeholders:** State and private engineering universities offering Telecommunication, Electronic, and Electrical Engineering degree programs (such as the University of Moratuwa and the University of Peradeniya). This also includes technical, vocational, and technological colleges where communication systems laboratories require practical instrumentation benches.
- **Secondary Stakeholders:** Academic lecturers and instructors who require real time demonstrative teaching aids, undergraduate students seeking intuition beyond math equations, and lab coordinators managing tight equipment procurement budgets.

1.5 Stakeholder Insights & Expert Consultation

Detailed consultation loops were conducted with Dr. Samiru Gayan, Senior Lecturer within the Department of Electronic & Telecommunication Engineering at the University of Moratuwa. Dr. Gayan confirmed that current university laboratory exercises are structurally skewed toward abstract software models. He emphasized that the introduction of a localized hardware development platform would resolve critical learning bottlenecks by providing immediate visual indicators for bitstream formation, packet boundaries, redundancy additions, and hardware packet rejection behaviors.

1.6 Project Objectives

The system targets the following core engineering goals:

- Construct a robust, standalone laboratory instrument that processes a standard data alphabet set ($S = \{A, B, C, D, E\}$) under user specified statistical profiles.
- Integrate a continuous software execution pipeline comprising interchangeable algorithmic blocks for Source Coding, Packetization framing, Channel Error Protection, and Carrier Modulation.
- Implement a dual display SPI processing topology utilizing STM32 DMA controllers to separate operational menu entry tracking from physical wave shape and constellation scatter layouts.
- Deploy a tactile hardware subsystem utilizing automated logic registers and analog noise pots to show bit error rate transformations dynamically.

1.7 System Scope & Boundaries

The console serves strictly as a baseband and digital carrier visualization instrument for academic laboratories. The hardware boundary focuses on structural logic validation and display rendering rather than high power RF transmission over antennae arrays. All signal mutations, carrier shapes, and channel path interference distributions are mapped directly within the core microcontroller logic engine and translated to dedicated visual interfaces.

2 System Architecture & Engineering Alternatives

2.1 Global Block Interconnect Architecture

The structural framework of the system relies on a modular, decoupled signal routing layout. High level compute blocks pass operational parameters directly to localized peripheral hardware layers.

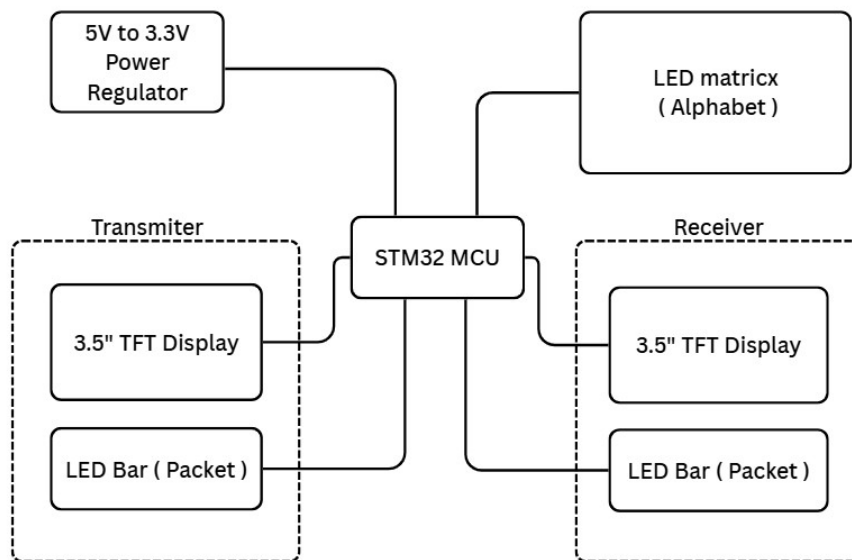


Figure 1: Global Block Interconnect Architecture and STM32 Bus Distribution

Figure 2: Enter Caption

2.2 Analysis of System Engineering Alternatives

During the conceptualization phase, three architectural implementations were evaluated. Table 1 summarizes the comparative trade offs that informed the final hardware selection.

Table 1: Design Matrix of Architectural Implementation Alternatives

Implementation Path	Engineering Advantages	Technical Disadvantages
Pure Software Simulation (MATLAB / Python)	High mathematical precision; easy implementation architectures; zero physical development limitations.	No tactile user interactions; completely separated from physical clock domains or hardware timing limits; zero real time interaction loops.
FPGA Hardware Platform	Parallel processing; sub-nanosecond clock matching; native signal generation speed.	High development overhead; complex HDL debug tracks; long compile cycles; expensive development boards.
STM32 ARM Cortex Micro-controller	Hardware level DMA, native nesting vector interrupts (NVIC), rich configuration of independent SPI peripherals, predictable real-time execution.	Finite parallel graphics rendering lanes compared to FPGAs (mitigated by optimized DMA display buffering).

3 Detailed Requirement Analysis & System Specifications

3.1 Functional Requirements

The hardware and software design layout must strictly execute five sequential signal operations.

1. **Tactile User Input Mapping:** Monitor an integrated physical matrix keypad to capture character strings from the asset alphabet ($S = \{A, B, C, D, E\}$), and capture user defined probability values through an interactive menu.
2. **Source Compression Compiling:** Run real time binary string generation using 3-bit Fixed Length formats, Variable Length Huffman encoding structures, and Shannon Fano splitting trees.
3. **Dynamic Packetization Framing:** Segment structural source bitstreams into distinct, user selectable frame boundaries.
4. **Channel Protection Processing:** Generate and inject mathematical parity configurations (Even/Odd options), Cyclic Redundancy Check (CRC-3/CRC-4 via 3-bit or 4-bit user-selected generator inputs)
5. **Modulation Phase Mapping:** Map generated bitstreams into discrete complex carrier vectors supporting Amplitude Shift Keying (ASK), Binary Phase Shift Key-

ing (BPSK), Quadrature Phase Shift Keying (QPSK), and 16-Quadrature Amplitude Modulation (16-QAM).

3.2 Non-Functional Requirements

- **UI Latency Limitations:** Frame updates across the dual-display TFT panels must sustain a minimum target of 30 frames per second (FPS), ensuring real-time responsiveness when modulating carrier graphs.
- **Structural Durability:** The mechanical enclosure must safely support continuous laboratory use inside university environments without thermal throttling or loose electrical junctions.
- **User Accessibility:** The interface layout must be intuitive for engineering undergraduates, featuring immediate menu execution links and physical control dials.

3.3 System Specifications Matrix

Table 2 lists the functional components and hardware interfaces used in the design.

Table 2: System Hardware and Architectural Component Specifications

System Parameter Domain	Target Implementation Element
Central Microcontroller	STM32 ARM Cortex-M Series (32-bit Architecture)
Primary Graphical Panels	2×3.5" SPI Thin-Film-Transistor (TFT) Displays (320×480)
Bitstream Mapping Displays	3× 74HC595 Serial-in Parallel-out Shift Register LED Arrays
User Input Device	4 × 4 Matrix Tactile Keypad
Channel Interference Control	1× Analog Linear Rotational Potentiometer
Power Supply Rail	Integrated 5V to 3.3V Low Dropout (LDO) Regulator Infrastructure

4 Hardware Realization & Electronic Circuit Design

4.1 Central Microcontroller Infrastructure (STM32)

The processing topology leverages the nested hardware vector architecture of an STM32 microchip. Processing routines are timed explicitly using an integrated Real-Time Operating System (RTOS) or bare-metal interrupt scheduling:

- **Core Communication Pipeline:** Executes keypad matrix debounce scans, state-machine switches, mathematical code-tree processing, packet framing isolation, and phase coordinate generation.
- **Graphics Multi-Buffering Engine:** Offloads frame-buffer arrays via dedicated STM32 DMA streams to the SPI lines, allowing zero-overhead pixel rendering.

4.2 Dual-Display SPI Interfacing Logic

To present operational menus alongside real-time carrier waveforms without causing bus contention or screen tearing, the display layout utilizes separate hardware SPI blocks (e.g., SPI1 and SPI2) on the STM32.

4.3 Serial Shift Register Bit-Array LED Monitoring Matrix

To display high-density binary streams at discrete processing points (Source output vs. Channel output) without consuming scarce MCU GPIO pins, three 74HC595 shift registers are cascaded. This allows the system to update 30 unique LED indicators using just 3 processor pins via synchronous latch clocked bursts.

4.4 Analog Channel Noise Simulation Circuit

The analog channel emulation stage maps physical user adjustments directly to digital noise generation models. A linear potentiometer provides an adjustable voltage vector across an internal STM32 12-bit ADC channel. The measured value sets the standard deviation σ within the firmware noise algorithm, driving an adjustable Bit Error Rate (BER) output layout:

$$V_{\text{ADC_Input}} = \left(\frac{R_{\text{pot_wiper}}}{R_{\text{total_track}}} \right) \cdot V_{\text{DD.3.3}} \quad (1)$$

5 Mathematical Formulations & Algorithmic Design

5.1 Source Coding Frameworks

The asset alphabet processing engine tracks input structures across a standardized symbol library:

$$S = \{A, B, C, D, E\} \quad \text{subject to} \quad \sum_{i \in S} P(i) = 1.0 \quad (2)$$

Huffman Coding Engine

The compression module builds a dynamic binary tree sorted by descending token weights. Codewords are generated recursively by tracing binary paths from the root node down to the individual character leaves.

Shannon-Fano Compression Logic

The alphabet tokens are sorted by descending probability and bisected into matching subsets to equalize the cumulative probability densities across each branch. A '0' binary bit is assigned to the upper partition and a '1' binary bit to the lower partition. This logic repeats recursively until every leaf node is isolated.

5.2 Channel Coding Verification Matrix

Cyclic Redundancy Check (CRC-3 / CRC-4)

The cyclic division block utilizes a software shift register verification framework. The system divides the expanded message block $X^n \cdot M(X)$ by a user-selected generator polynomial $G(X)$ (supporting both 3-bit and 4-bit polynomial inputs configured via software menu loops) using bitwise XOR operations to determine the trailing validation bits $R(X)$:

$$R(X) = \text{Remainder} \left[\frac{X^n \cdot M(X)}{G(X)} \right] \quad (3)$$

Linear Block Hamming Code Protection

The system implements a $(7, 4)$ systematic Hamming single-error correction matrix. Message nibbles are mapped to standard block positions, and parity elements are computed at power-of-two index coordinates $(1, 2, 4)$. At the receiver block, error vectors are isolated via matrix syndrome checks:

$$\mathbf{S} = \mathbf{H} \cdot \mathbf{Y}^T \quad (4)$$

If the calculated syndrome vector $\mathbf{S} \neq \mathbf{0}$, the system pinpoints the flipped bit position, corrects it automatically, and triggers a localized indicator LED on the console.

5.3 Packet Validation and Rejection Mechanics

When packet structures run with injected noise anomalies, the receiver subsystem conducts automated parity check comparisons and structural CRC recalculations. If the computed remainder fields or parity sums yield alignment failures against the transmitted payload headers, the active TFT display marks the corresponding packet frame as ****Rejected****, discarding the corrupted array and incrementing an onboard real-time dropped-packet tally.

5.4 Digital Carrier Modulation Phase-Mapping

The digital mapping core transforms binary frames into discrete, high-frequency mathematical vectors in phase-space:

$$s_{\text{BPSK}}(t) = A \cos(2\pi f_c t + \pi \cdot b_k) \quad b_k \in \{0, 1\} \quad (5)$$

$$s_{\text{QPSK}}(t) = A \cos\left(2\pi f_c t + \frac{(2m+1)\pi}{4}\right) \quad m \in \{0, 1, 2, 3\} \quad (6)$$

$$s_{\text{16QAM}}(t) = I_k \cos(2\pi f_c t) - Q_k \sin(2\pi f_c t) \quad I_k, Q_k \in \{-3, -1, 1, 3\} \quad (7)$$

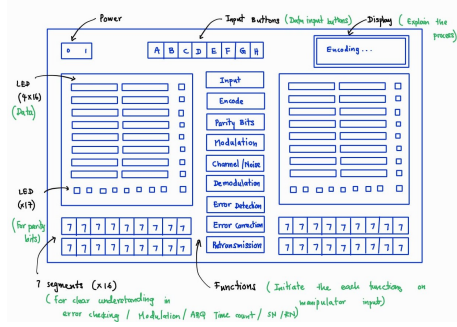
6 Enclosure Architecture & Mechanical Design

6.1 Design Constraints & Material Selection

The mechanical enclosure balances long term structural stability. The high impact Polylactic Acid (PLA) polymers were chosen for the body shell to minimize total tool weight while providing adequate structural rigidity. We use a optimal infill rate to achieve the optimal balance between cost and strength. The clear structural windows are integrated across the display faces to preserve access to tactile controls. The heat management considerations were also integrated directly into the spatial layout to ensure adequate passive cooling for the internal power regulation circuitry.

6.2 Mechanical CAD Design Evolution

The mechanical architecture transitioned through multiple parametric CAD iterations, driven primarily by a philosophy of human centric design to prioritize user interaction and desktop ergonomics. The display mount angles were specifically re-engineered to provide an optimized, reflection free viewing axis for collaborating laboratory teams. Beyond interface ergonomics, the geometric evolution carefully balanced physical stability on the workbench, a professional aesthetic appearance of a laboratory equipment, and strict material cost constraints, ensuring the final enclosure is both highly functional and economical to fabricate.



(a) Initial Layout (Sketched Layout)



(b) Finalized CAD Render Model

Figure 3: Enclosure Mechanical Evolution and Interface Formatting Layout

6.3 Professional CAD Methodologies & Skill Application

The mechanical development phase was executed using SolidWorks, an industry standard 3D CAD software suite. To ensure dimensional coherence across all mating components, a top down assembly modeling approach was adopted. This methodology allowed the enclosure to be conceptualized and scaled as a single, unified structure before being partitioned into distinct manufacturable bodies.

To maximize design flexibility and reflect professional engineering practices, the model was constructed using equation driven parametric dimensions governed by a master configuration of global variables. This ensured that any adjustments to display clearances or internal PCB footprints automatically cascaded through the entire mechanical assembly. Furthermore, advanced CAD tools were utilized to implement robust structural elements, specialized internal mounting bosses to securely anchor the hardware.

7 PCB Layout Architecture & Isolation Strategies

7.1 Electrical Schematic Design Topology

The electronic design uses a modular hardware topology to improve system reliability. The physical layout is divided into two primary printed circuit boards (PCBs).

The first board is a dedicated power supply PCB. This board handles voltage regulation and safely distributes stable power to the rest of the system. The second board is the centralized MCU PCB. It hosts the main STM32 microcontroller and acts as the brain of the console.

All external input and output peripherals, such as the TFT displays and the keypad, connect directly to this central MCU board. We utilize standard JST connectors for these peripheral links. This modular approach makes the physical assembly much easier. Furthermore, it cleanly isolates the sensitive high-speed digital data paths from the main power circuits.



Figure 4: Central STM32 System Schematic and Bus Track Routing Breakouts

7.2 Power Supply & Voltage Regulation PCB

The system receives a standard 12V DC input from an external laboratory power supply dock. To handle this input efficiently, we developed a dedicated power distribution PCB. Instead of using traditional linear regulators, this board utilizes switching mode buck converter ICs to maximize power efficiency and reduce heat generation.

We designed two separate step down conversion circuits on this board. The first buck converter drops the 12V input down to a stable 5V rail. This 5V supply is exclusively routed to drive the power hungry components, specifically the dual TFT displays and the LED monitoring arrays. The second buck converter steps the voltage down to a clean 3.3V rail. This 3.3V supply safely powers the main STM32 microcontroller and the remaining digital logic circuits.

By utilizing switching mode ICs for both conversion stages, the system achieves significantly higher overall power efficiency (more than 85%). This ensures the console remains cool and stable during continuous laboratory operation.



Figure 5: 12V to 5V and 3.3V Dual Buck Converter Power Schematic

7.3 PCB Layer Optimization & Signal Integrity

We fabricated the main board using a standard double sided FR4 printed circuit board. The board features a 1.6 mm thickness and a 1 oz copper weight.

To maintain strict signal integrity, we completely isolated the digital and analog routing paths. All high-speed SPI data traces are routed exclusively on the top copper layer. Conversely, the sensitive analog routing for the noise simulation is restricted to the bottom layer. This physical separation prevents digital crosstalk from corrupting the analog signals.

Furthermore, we implemented a solid, unbroken ground plane across the bottom layer. This extensive ground pour provides a low impedance return path for all circuits. It is highly effective at preventing ground loops and minimizing high frequency electromagnetic interference (EMI).

8 Testing, Verification & Experimental Calibration Results

8.1 Algorithmic Verification Parameters

To evaluate the compression engine, variable-length source profiles were validated using a standard test sequence string (*AABACAE*). The experimental performance results are summarized in Table 3.

Table 3: Algorithmic Compression and Code-Word Length Test Matrix

Input Profile Vector	Coding Mode	Resulting Bits	Entropy Factor
<i>AABACAE</i>	Fixed-Length Base	21 Bits	1.0000
<i>AABACAE</i>	Huffman Engine	13 Bits	0.6190
<i>AABACAE</i>	Shannon-Fano Mode	14 Bits	0.6667

8.2 Potentiometer Noise Calibration and Signal Profile Mapping

The analog noise emulation stage was calibrated using an external digital storage oscilloscope. The relationship between physical control dial position, raw internal 12-bit STM32 ADC register tracking values (0 to 4095), and the generated peak-to-peak voltage noise envelope is documented in Table 4.

Table 4: STM32 12-bit ADC Noise Mapping Calibration Data

Rotational Angle ($^{\circ}$)	Measured 12-bit ADC Register	Calculated Output σ Voltage (V_{p-p})
0 (Minimum Limit)	0x000	0.00 V
45	0x249	0.15 V
90	0x492	0.32 V
135	0x6DB	0.48 V
180 (Midpoint)	0x924	0.65 V
225	0xB6D	0.82 V
270 (Maximum Limit)	0xFFF	1.20 V

9 Financial Procurement Analysis & Scaled Cost Budgeting

9.1 Prototyping Bill of Materials (BOM)

This section details the hardware expenditure required to build the initial laboratory prototype of the console. The total development cost reflects small volume retail pricing for individual components rather than bulk rates. Table 5 provides a complete itemized breakdown of these expenses.

Table 5: Itemized Component Budget for the Initial Prototype

Component Description	Unit Price (LKR)	No. of Units	Final Cost (LKR)
STM32 ARM Cortex MCU	2,200	1	2,200
3.5" SPI TFT Graphic Display Modules	2,750	2	5,500
74HC595 Logic Shift Registers	50	3	150
High-Intensity Indicator LEDs	3	25	75
LED Light Bars	50	4	200
4×4 Matrix Tactile Keypad	1,200	1	1,200
Linear Analog Potentiometer Dial	50	1	50
Double-Sided FR4 Custom PCB Fabrication	3,500	1	3,500
Custom 3D Printed Mechanical Enclosure Shell	23,000	1	23,000
Passive Support Components (LCSC)	3,500	1	3,500
Logistics	2,450	1	2,450
12V Power Adapter	1,775	1	1,775
Total Prototyping Budget			LKR 43,600

9.2 Mass Production & Scaled Cost Budgeting

This section forecasts the unit cost when scaling production to an estimated 100 to 200 units per year. Sourcing components in bulk directly cuts the individual component prices.

First, the custom PCB fabrication cost drops significantly because the initial setup fees are divided across many boards. Second, importing electronic components in bulk batches drastically reduces the unit purchasing rates and shipping overhead. Finally, shifting from single unit 3D printing to low-volume injection molding reduces the enclosure cost once the initial mold cost is distributed across the production run. Table 6 summarizes these optimized manufacturing costs per unit.

Table 6: Estimated Unit Budget for Scaled Production (100–200 Units/Year)

Component Description	Unit Price (LKR)	No. of Units	Final Cost (LKR)
STM32 ARM Cortex MCU Module	1,600	1	1,600
3.5" SPI TFT Graphic Display Modules	2,350	2	4,700
4×4 Matrix Tactile Keypad	850	1	850
Double-Sided FR4 Custom PCB Fabrication	600	1	600
Scaled Injection-Molded Mechanical Enclosure	9,900	1	9,900
Passive Support Components (LCSC)	3,150	1	3,150
12V Power Adapter	975	1	975
Total Scaled Unit Budget			LKR 21,775

9.3 Final Production Cost

The raw component budget does not reflect the complete price of the product. To calculate the true final production cost per unit, we must include several manufacturing overheads.

The following additional costs are considered for the mass production phase.

- **Assembly and Labor Costs (LKR 450):** Expenses for automated PCB assembly (Pick-and-Place) and manual soldering of larger components.
- **Testing and Quality Assurance (LKR 950):** The cost of functional testing, hardware calibration, and defect inspection for each unit.
- **Packaging and Logistics (LKR 600):** Expenses for protective foam packaging, printed user manuals, and institutional shipping.
- **Legal and Certification Fees (LKR 450):** The distributed costs for safety certifications, university tender documentation, and intellectual property filings.
- **Warranty and Support Buffer (LKR 1200):** A small reserve fund allocated per unit to cover potential hardware replacements or future technical support.

By adding these manufacturing overheads (LKR 3,650) to our scaled unit component budget (LKR 21,775), we determine the final market-ready cost.

Final Mass Production Cost Per Unit: LKR 25,425

10 Business Model & Market Viability

10.1 Market Analysis & Targeted Implementations

Commercial telecommunication laboratory instrumentation platforms are typically imported from international markets, with procurement costs often reaching several million LKR per unit. This pricing model creates significant budgetary constraints for public and private engineering faculties in developing economies. The console bridges this gap by targeting:

- Government and private university electronic and telecommunication engineering laboratories.
- Regional technical colleges and vocational training institutes.
- Advanced secondary school STEM training facilities.

10.2 Unique Value Proposition

- Integrated Multi-Stage Pipeline: Combines source encoding, packetization constraints, channel error correction, and high order modulation schemes into a single standalone platform.
- Tacitly Immersed Interface: Replaces purely virtual software environments with physical potentiometer-driven noise injectors and diagnostic monitoring LEDs.
- Significant Cost Advantage: Delivers a standalone laboratory instrument at a production price point that enables higher student-to-bench density ratios.

10.3 Commercialization and Sales Strategy

Our commercial strategy focuses on clear revenue streams and accessible customer channels.

- **Revenue Model:** Our primary profit earning method is direct sales. We are mainly focusing on the Business-to-Business (B2B) market, targeting universities and technical institutes. However, we are also open to the Business-to-Consumer (B2C) market for individual engineering students and hobbyists.

- **Primary Online Portal:** We will develop a dedicated official website. This website will act as the main communication bridge between our team and the customers. It will handle direct order placements, customer support, and digital user manuals.
- **Local Retail Partnerships:** To increase our physical market presence, we plan to collaborate with well-known electronic equipment shops across Sri Lanka. These local retail stores will accept customer orders and help us reach a wider audience.
- **Pricing Strategy:** Based on our final mass production cost of LKR 25,425, we can set a competitive market price. A suggested retail price of LKR 59,000 to LKR 65,000 provides a strong profit margin of 135% while remaining highly affordable for local engineering faculties.

11 Future Technical Expansion Tracks

11.1 Hardware-Defined Radio Sub-System Mixers

Future iterations will introduce an optional analog RF up-conversion stage. Integrating a high-speed Software-Defined Radio (SDR) I/Q mixer sub-board would allow the console's digital baseband modulation waveforms to be transmitted over physical ISM radio frequencies (e.g., 2.4 GHz), creating an end-to-end physical wireless testing platform.

11.2 Real-Time Bit Error Rate (BER) Graphical Plotting Engine

An upcoming software update will introduce an automated, long-term testing routine. This module will systematically adjust the analog noise floor to construct and render real-time Waterfall and BER vs. E_b/N_0 curves directly onto the auxiliary graphics display.

12 Project Conclusion

The *Digital Communication Visualization Console* successfully meets the technical design requirements established for module EN2160. By implementing source compression models, packet framing constraints, block error control coding, and high-order phase modulation on an optimized STM32 architecture, Team Synex has developed a robust educational laboratory instrument. This platform moves beyond static theoretical lectures and complex FPGA programming, providing engineering undergraduates with a tactile, real-time tool to bridge theory and practice. The production, cost, and design analyses demonstrate high implementation readiness for integration into undergraduate laboratories at the University of Moratuwa.

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